

Kaisheng Liu

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Education

University of Toronto

Expected November 2026

Major in Computer Science and Statistics

Toronto, Ontario

- Arts and Science Internship Program (ASIP)
- Relevant Coursework: Software Tools and System Programming, Data Structures and Algorithms Analysis (Python & Java), Neural Network and Deep Learning, Computer Organization, Database Design (PostgreSQL), Operating System, Software Design

Technical Skills

Programming Languages: Python, Java, C, Javascript/Typescript, SQL, R, Assembly

Frameworks & Tools: React, Node.js, PyTorch, Database (MySQL, PostgreSQL, Firebase, MongoDB), Tableau, JUnit, Git, PyUnit, YOLOv8, OpenCV

Experience

BalanceAI Educational Language Assessment Lab

Sept 2024 – April 2025

OISE Education Program Work Study

Toronto, ON

- Designed gamified and streak system by using **React** and **TypeScript** that implements a specific rewarding feedback from learning, enhancing user engagement.
- Optimized the authentication procedure by implementing advanced **licensing and data encryption** techniques, ensuring system security.
- Implemented computational algorithm to assess the literacy results into aggregate data dashboard with fast generating result speed
- Redesigned state management system in **Firebase**, reducing loading times for scores and tasks by **2 seconds** and improving scalability.

Yisheng Pharmacy Company

May 2023 – August 2023

Software Engineering Intern

Shanghai, China

- Collaborated with the software development team to design and implement solutions supporting various business operations.
- Developed and optimized SQL queries for efficient database management and retrieval of customer health-related indexes, improving query performance by 20%.
- Contributed to the development of interactive dashboards using Tableau, enhancing data-driven decision-making capabilities.
- Built Python scripts for automation, data processing, and backend integration, reducing manual effort by 30%.

Projects

Eventure | Java, Swings, Architecture, Unit Test

- Designed and Implemented a task management application utilizing **Java** and **JavaScript**, allowing users to chat, assign, and track events, and view profile.
- Led a team of six in conceptualizing the theme, designing the structure, and developing the software, ensuring effective communication and reasonable task allocation
- Conducted comprehensive unit tests for use cases using JUnit, achieving **86%** test coverage, ensuring robust application functionality and reliability.

NorthStar Fortune Insights – AI Guidance Web App | TypeScript, React, Next.js, Node.js, PostgreSQL, Redis, OpenAI API, Docker

- Developed a full-stack guidance platform using **TypeScript**, **React/Next.js**, **Node.js**, and the **OpenAI API** to turn user questions into structured, explainable life and career recommendations via a rules engine + **LLM** pipeline.
- Architected typed **RESTful APIs** and relational models in **PostgreSQL**, added **Redis** caching, and containerized services with **Docker** to improve latency and simplify deployment.
- Implemented a retrieval-augmented generation (RAG) workflow and product analytics to power a freemium + subscription model tailored for Canadian and North American users.

Sports Field System Analysis | *Python, Computer Vision, YOLO, Machine Learning*

- Implemented **YOLOv8** for object detection of players, referees, and footballs in video footage.
- Developed a t-shirt color-based team assignment system using KMeans clustering for pixel segmentation.
- Applied perspective transformation with **OpenCV** to compute movement in real-world units and integrated real-time object tracking across frames for continuous player analysis.
- Measured camera movement using optical flow techniques to ensure accurate player movement analysis.